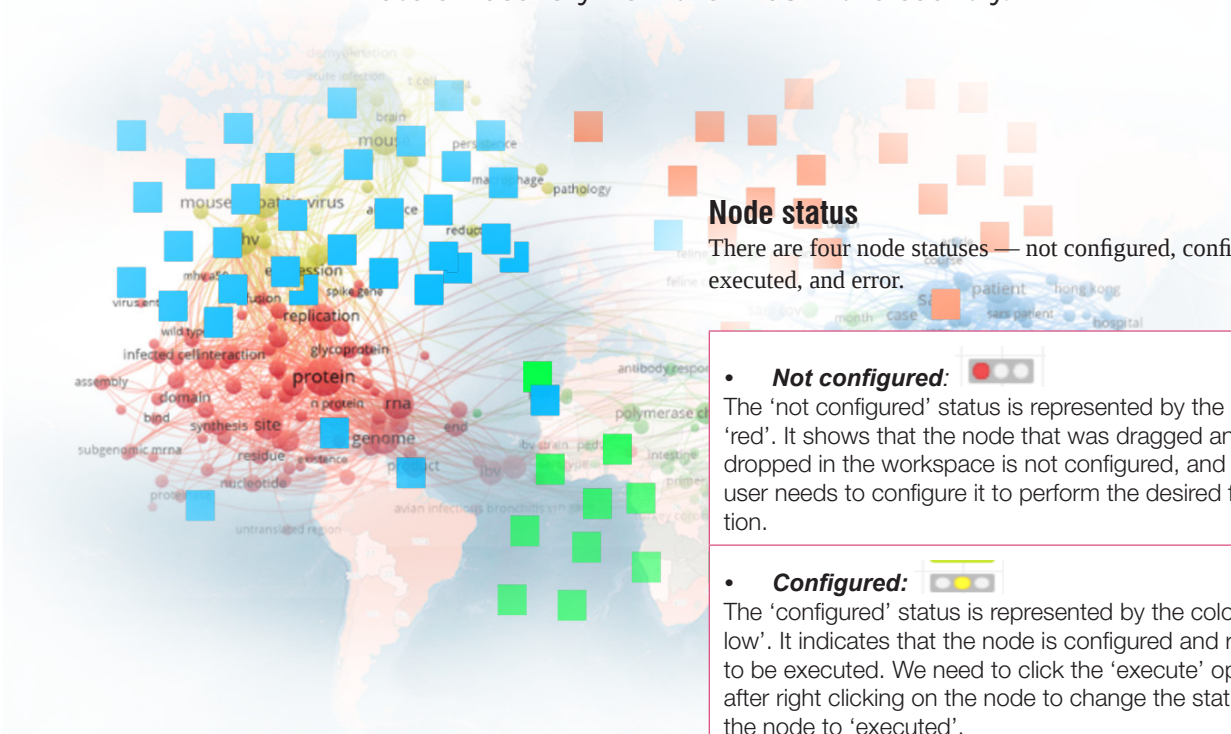
Using KNIME to Understand the Impact of Covid 19

KNIME (Konstanz Information Miner) is free, open source software being used in data science. It makes understanding data, and designing data science workflows and reusable components, easy for everyone. This article implements a clustering algorithm through KNIME for the Covid 19 data set to assess the rate of recovery from the virus in the country.



With KNIME, one can create visual workflows for data analytics with an intuitive, drag and drop style graphical interface, without the need for coding. KNIME also makes blending data from any source easy. In this article, we will be using the Covid 19 data set. Three clusters have been created based on the confirmed Covid-positive tests, number of people cured (recovered), and deaths as of date. The k-means node is used to perform k-means clustering on the data set. The colour manager and shape manager nodes are used to assign colours and shapes to the clusters. The scatter plot node is used to view the clusters as scatter plots.

To install KNIME:

- Download the KNIME Desktop Application from *https://www.knime.com/*.
- Install the KNIME Desktop Application and provide the necessary permissions during the setup process.

Node status

There are four node statuses — not configured, configured, executed, and error.

- ***Not configured:***

The 'not configured' status is represented by the colour 'red'. It shows that the node that was dragged and dropped in the workspace is not configured, and the user needs to configure it to perform the desired function.

- ***Configured:***

The 'configured' status is represented by the colour 'yellow'. It indicates that the node is configured and ready to be executed. We need to click the 'execute' option after right clicking on the node to change the status of the node to 'executed'.

Executed:

The 'executed' status is represented by the colour 'green'. It indicates that the node is successfully executed. If the execution fails, the user will see an 'error' status.

Error:

The 'error' status is represented with a cross inside a red circle. It indicates that the node has not executed properly or there is an error while configuring it.

Clustering

K-means clustering is an unsupervised machine learning iterative algorithm, which uses centroid-based selection. This algorithm places 'k' cluster centres and centroids randomly for each data point attached to it as the nearest cluster centre. This turns the centroids into cluster centres.